CIS 517: Data Mining and Warehousing
Project Report Template

Section:
Project Group Number (As in Blackboard):
Submission Date:
Instructor:

Members:

No.	Student Name	Student ID	Assigned Contribution	Related Section(s)
1				
2				
3				
4				
5				
6				
7				

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1. Dataset Overview and Project Objectives

Introduce the Online Shoppers Purchasing Intention Dataset, state the classification objective, describe the target variable, and summarize the dataset dimensions and attribute types. Do not copy the dataset description verbatim; explain it in your own words.

2. Python Environment and Packages

Identify and correctly use the Python packages required for analysis and modeling. Explain the purpose of the important packages and key code used.

2.1. Packages Used

Table 1: Packages Used

Package / Library	Purpose in This Project	Example Use

3. Statistical Evaluation and Data Understanding

Use suitable statistical methods to understand the dataset and its attributes. Explain what each result means rather than only showing output.

4. Data Preprocessing

Prepare the data for modeling and explain why each preprocessing step is needed. The report should make the sequence of preprocessing steps clear.

5. Outlier Analysis

Identify outliers using an appropriate method, evaluate whether they are valid observations or problematic values, and explain how they were handled.

6. Classification Models and Performance Tuning

Apply Logistic Regression, Decision Tree, and Random Forest. Show model setup, tuning/performance enhancement, and evaluation.

6.1 Logistic Regression

For this model, include: model setup, important parameters, tuning or performance enhancement, evaluation results, and a short interpretation.

6.2 Decision Tree

For this model, include: model setup, important parameters, tuning or performance enhancement, evaluation results, and a short interpretation.

6.3 Random Forest

For this model, include: model setup, important parameters, tuning or performance enhancement, evaluation results, and a short interpretation.

7. Model Performance Comparison

Compare the performance of the models using suitable evaluation measures learned. The comparison should be meaningful and use the same evaluation basis for all models.

8. Discussion

Discuss the main findings of the project. Connect the preprocessing choices, statistical observations, model behavior, and evaluation results. Mention relevant limitations of your analysis.

9. Conclusion

Summarize what you learned from the analysis and modeling. Keep the conclusion focused on findings already supported by the project results.

References

List all sources used in the project, including datasets, books, journal articles, websites, documentation, and other relevant materials. Use a consistent academic referencing style throughout the report. All sources cited in the report must appear in this section.

Appendix A. AI Usage Declaration and Prompt Log

Required if AI was used at any stage of the project. *Declare the AI tool and purpose, and include every prompt used. Each prompt must be linked to the relevant project task or report section. If no AI was used, state that clearly below.*

AI Usage Declaration:

[ ] No AI tools were used in this project.
[ ] AI tools were used and are fully declared below.

AI Usage Declaration Form:

Table 2: AI usage declaration

AI Tool	Purpose	Related Task / Section	Exact Prompt Used	How the Output Was Used / Verified
Example: ChatGPT	*Grammar polishing*	*Section 5.2*	*Improve the grammar of the following paragraph without changing its meaning: ...*	*Language only; final wording reviewed by the team.*

Important: The appendix should contain the prompts used, not a general description such as "used AI for help."

Appendix B. Plagiarism Report

Attach the plagiarism report generated for the final submitted report. The similarity percentage must be below the required threshold specified in the project guidelines.

Appendix C. Supplementary Code, Tables, or Figures (Optional)

Use this appendix only for supporting material that is useful but too detailed for the main report, such as longer code extracts, additional evaluation tables, or supplementary figures. The main report should still contain the key results and explanations.